# Dani Roytburg

CURRICULUM VITAE

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Master's student in the Machine Learning Department at Carnegie Mellon University focused on reliable and interpretable AI, multi-agent training paradigms, and activation-based model analysis.

## Education

**Carnegie Mellon University, School of Computer Science** expected May 2027

M.S. in Machine Learning

**Emory University, College of Arts and Science** 2025

B.S. in Computer Science; Quantitative and Social Sciences · *summa cum laude*

## Research & Professional Experience

**ML Alignment and Theory Scholars** May 2026 – September 2026

Research Fellow · Berkeley, CA

Advised by Shi Feng. Developed persona-training methods which emulate midtraining for better model organisms and constitutions. Engineered evaluations across 100+ training runs on synthetic-document pipelines. Evaluated how tastefully frontier models can test ideas in safety research.

**Carnegie Mellon University** September 2025 – Present

Research Assistant, DI Group · Pittsburgh, PA

Develop multi-agent reinforcement learning environments to improve legibility of reasoning traces. Design user studies and adversarial setups to measure influence of legibility on human and multi-agent interactions.

**Martian Research** May 2025 – February 2026

Research Fellow · San Francisco, CA

Engineered activation profiles of language model evaluators to suppress self-preferential bias, applied to LLM routers (ICML 2026, MechInterp @ NeurIPS 2025). Winner of Mechanistic Router Interpretability Hackathon (40 submissions).

**Emory University** November 2021 – May 2025

Research Assistant, Digital Humanities Lab · Atlanta, GA

Designed and trained large-scale graph-text models; evaluated adversarial robustness of LLMs; built historical correspondence networks and dashboards.

**ZS Associates** June 2024 – August 2024

Internship - Business Technology Solutions · Evanston, IL

Automated ETL pipelines and implemented synthetic-data imputation for compromised client data.

**Cloverpop** January 2024 – May 2024

Intern — Machine Learning · Chicago, IL

Deployed decision-management systems and structured-prediction models on meeting transcripts.

**McCormick and Company** June 2023 – August 2023

Intern — Global Marketing and Sales Data Science · Hunt Valley, MD

Built Amazon sales data visualizations and automated analytics with GCP BigQuery and BERT-based embedding search.

**Northeastern University** June 2022 – October 2022

Research Assistant, Mediacloud Lab · Boston, MA

Extended NER to long-tail ethnics cuisines using Bon Appetit data.

## Publications

* Equal contribution / co-first authorship.

### Under review

**Pre-training interventions, ex post facto: Grafting model beliefs across checkpoints**
Peter Nutter*, **Dani Roytburg***, Clément Dumas, Jinghua Ou, Shi Feng.
*Under review*, 2026.
Paper · Blog post · Project website · Code

### Conference & workshop papers

**Disentangling Models from Personas in Heterogeneous LLM Simulations**
**Dani Roytburg**, Daphne Ippolito.
*Second Workshop on Social Simulation with LLMs: Fidelity in Applications* (*COLM 2026, Spotlight, Best Paper Award*), 2026.
Paper · Project website · Code

**Measuring Weak-to-Strong Legibility of Reasoning Models**
**Dani Roytburg**, Shreya Sridhar, Daphne Ippolito.
*Advances in Neural Information Processing Systems* (*NeurIPS*), 2026.
Also at: ICML 2026 Workshop on AI for Good (AI4GOOD).
Paper

**Are LLM Evaluators Really Narcissists? Sanity Checking Self-Preference Evaluations**
**Dani Roytburg**, Matthew Bozoukov, Matthew Nguyen, Jou Barzdukas, Mackenzie Puig-Hall, Narmeen Oozeer.
*Forty-Third International Conference on Machine Learning*, 2026.
Paper · Code

**Mind the Gap: Pathways Towards Unifying AI Safety and Ethics Research**
**Dani Roytburg***, Beck Miller*.
*Proceedings of the International Association for Safe and Ethical AI*, 2026.
Paper · Code

**Breaking the Mirror: Activation-Based Mitigation of Self-Preference in LLM Evaluators**
**Dani Roytburg***, Matthew Bozoukov*, Hongyu Fu, Matthew Nguyen*, Jou Barzdukas*, Narmeen Fatimah Oozeer.
*Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025.
Also at: NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle; NeurIPS 2025 Workshop on Reliable ML from Unreliable Data.
Paper · Code

**Words and Action: Modeling Linguistic Leadership in # BlackLivesMatter Communities**
**Dani Roytburg***, Deborah Olorunisola*, Sandeep Soni, Lauren Klein.
*Proceedings of the International AAAI Conference on Web and Social Media*, 2025.
Paper

## Awards

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| Best Paper Award, COLM Social Simulation Workshop | 2026 |
| Best Poster, LTI Student Research Symposium | 2026 |
| Emory University Dean's List | 2022, 2024, 2025 |
| Winner, Martian Research Mechanistic Interpretability Hackathon | 2025 |

## Technical Skills

Python · Java · R · JavaScript · Typescript · SQL · PyTorch · JAX · HuggingFace Transformers · scikit-learn · spaCy · networkx · D3.js · React.js · GCP · Docker · MySQL